ProteinBERT: A protein sequence & function deep-learning model

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Tl;dr – A protein language model

1

- Self-supervised deep language modeling has shown success in natural language and protein tasks, but previous models were designed for natural language.
- ProteinBERT is specifically designed for proteins.
- We introduce a novel pretraining task (Gene Ontology (GO) annotation prediction)
- Efficient, flexible to very large sequence lengths, and interpretable thanks to global attention
- Excellent performance on diverse benchmarks, despite being far smaller model than competing language models methods
- Code (Keras + Pytorch) and pretrained model weights available: https://github.com/nadavbra/protein_bert

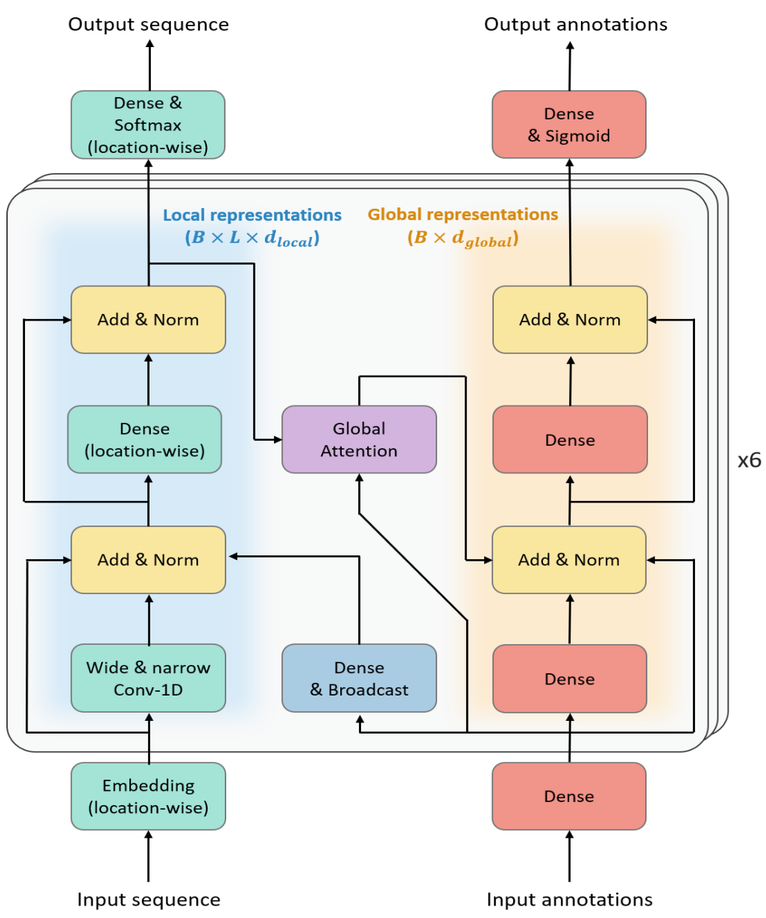
ProteinBERT at a glance

3

- Gene Ontology annotations pretraining task & supported input
- Denoising based pretraining
- Different model architecture
  - Faster, smaller than BERT
  - Linear, global attention
  - Flexible sequence lengths supported (>10k)
  - Mixed transformer/attention + convolutions
- Supports global and local (sequence) inputs and outputs
- Strong results on benchmarks despite size (just 16M params, 6 layers)

Model architecture

4



Unlike standard Transformers, ProteinBERT supports both local (sequential) and global data. The model has 6 transformer blocks manipulating local (left side) and global (right side) representations using fully-connected and convolutional layers, and residual connections. A global attention layer "connects" the local and global components, with linear complexity (as opposed to quadratic self-attention).

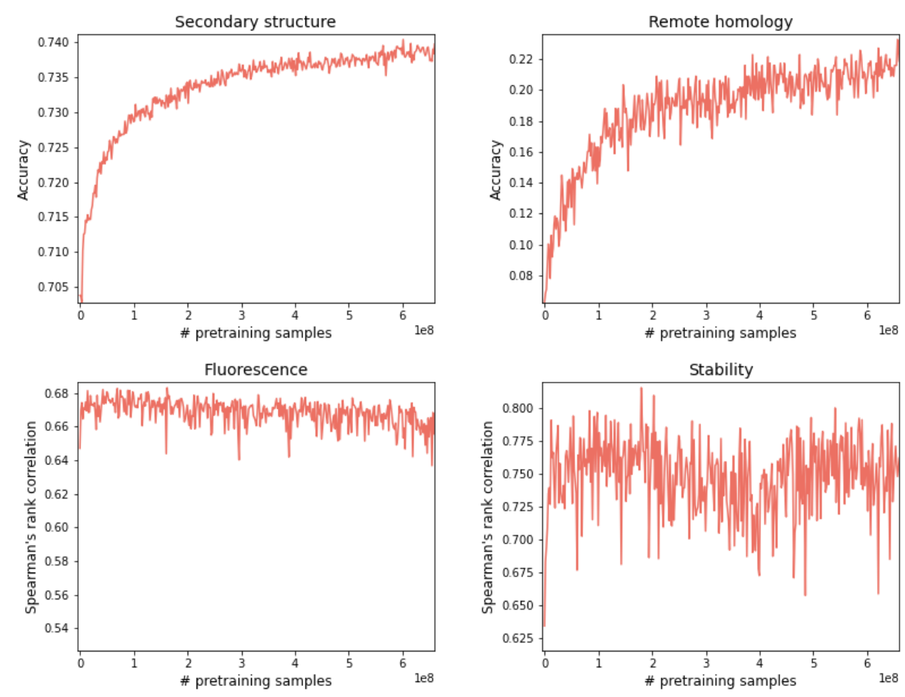
Pretraining Stats

2

- 106,000,000 UniRef90 proteins, with 8,943 possible Gene Ontology labels
- 43M proteins had any annotation, 2.3 annotations per protein.
- Proteins similar to any in benchmark test sets removed
- Trained for 1 month on 1 GPU: ~6.4 epochs, 670M sequences

Pretraining impact on benchmark tasks

5



Performance of fine-tuned ProteinBERT models over TAPE benchmarks test set as a function of pretraining amount

Understanding global attention

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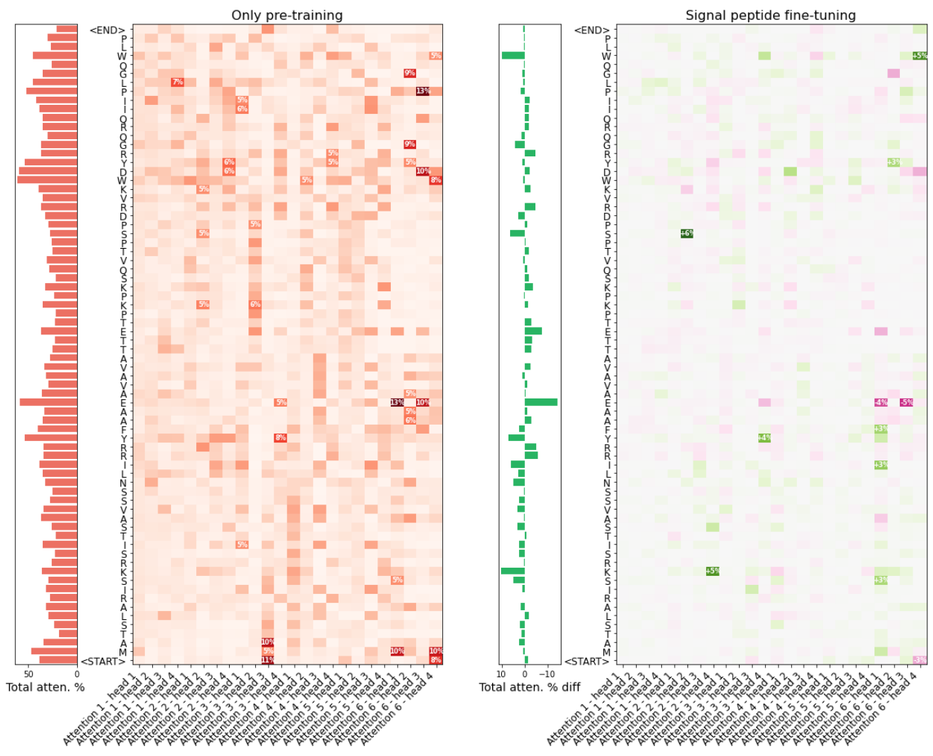

The heatmap of global attention values at each residue by each of the model's 24 attention heads. The bar plot shows the total attention at each residue by summing the attention values across all attention heads. The right panel shows the difference in attention values after fine-tuning ProteinBERT for signal peptide prediction, over GAMMACAL2

Conclusions

7

- **ProteinBERT's architecture is efficient and highly scalable, allowing it to process protein sequences of any length**
- **More pretraining and compute show clear performance gains (similar to other domains) – but only for some tasks**
- **GO annotations can improve on language modeling in proteins**
- **ProteinBERT has excellent performance on diverse tasks, despite it's size, especially with limited labeled data**

Publications:

ProteinBERT: A universal deep-learning model of protein sequence and function
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